

Vibrational Density-of-States Normalization Specification

VS3: Explicit VDOS Normalization of a Velocity Spectrum

mdstats

2026-07-22

Contents

1 Purpose and implementation stage	2
2 Terminology and interpretation	2
3 Provenance	2
3.1 Borrowed physical background	2
3.2 mdstats-specific contribution	2
4 Dependency boundary	2
5 Public function	3
6 Public result	3
7 Input contract	3
7.1 Source weighting	4
8 Frequency selection	4
9 Negative-value policy	4
10 Normalization modes	4
10.1 none	4
10.2 unit_area	4
10.3 degrees_of_freedom	4
11 Discrete normalization measure	5
12 Algorithm	5
13 Result identities	5
14 Edge cases and failure policy	6
15 Complexity and memory	6
16 Required tests	6
17 Deferred work	6
18 H0 signature propagation and deep immutability	7
19 References	7

1 Purpose and implementation stage

This document specifies stage VS3 in

`mdstats/analysis/velocity_spectrum.py`

The public function is

```
compute_vdos(spectrum, ...)
```

It converts an existing `VelocitySpectrumResult` into an explicitly normalized finite-temperature vibrational density of states. It does not recompute a VACF, Fourier transform a trajectory, infer phonon eigenmodes, calculate infrared or Raman intensities, or apply two-phase thermodynamics.

2 Terminology and interpretation

The result is called a **vibrational density of states** (VDOS). The more specific label **phonon density of states** is justified only for a crystalline or nearly harmonic solid where collective normal-mode language remains physically meaningful.

For liquids, molten salts, highly anharmonic solids, and diffusive ions, the same normalized function is a finite-temperature velocity-derived VDOS. Its low-frequency region may contain translation, diffusion, hopping, and cage motion in addition to oscillatory vibrations.

A VDOS is not an optical spectrum. Infrared intensity requires a dipole or charge-current correlation; Raman intensity requires a polarizability correlation.

3 Provenance

3.1 Borrowed physical background

Velocity-correlation spectra as descriptions of atomic motion are established in molecular dynamics, including Rahman's liquid-argon analysis [1]. The Wiener-Khinchin relation used by the source spectrum is attributed in the VS1 specification. Lin, Blanco, and Goddard later used a VACF-derived density of states in the two-phase thermodynamic method [2]. VS3 does **not** implement that method, its gas/solid decomposition, or any thermodynamic formula.

3.2 mdstats-specific contribution

The following are mdstats design decisions:

- separating the raw velocity spectrum from VDOS normalization;
- using the discrete one-sided FFT-bin measure rather than trapezoidal endpoint weights;
- requiring an explicit target for degrees-of-freedom normalization;
- refusing to infer removed constraints from atom count alone;
- applying one scalar factor to total, Cartesian, and per-atom projections;
- explicit low-frequency cropping without interpolation;
- strict material-negative rejection and roundoff-only clipping;
- the immutable result schema, validation identities, terminology metadata, and provenance payload.

4 Dependency boundary

VS3 consumes only

`VelocitySpectrumResult`

and the private helper

`spectral_bin_integral`

It does not depend on trajectory collections, velocity selection, VACF construction, or quadrature utilities.

5 Public function

```
def compute_vdos(
    spectrum: VelocitySpectrumResult,
    *,
    normalization: Literal[
        "unit_area",
        "degrees_of_freedom",
        "none",
    ] = "unit_area",
    target_degrees_of_freedom: float | None = None,
    minimum_frequency_thz: float | None = None,
    negative_policy: Literal["error", "clip_roundoff"] = "clip_roundoff",
    negative_tolerance: float = 1.0e-12,
) -> VDOSResult:
    ...
```

6 Public result

```
@dataclass(frozen=True, slots=True)
class VDOSResult:
    frequencies_thz: NDArray[np.float64]
    wavenumbers_cm_inv: NDArray[np.float64]
    energies_mev: NDArray[np.float64]

    total: NDArray[np.float64] # (F,)
    components: NDArray[np.float64] # (F, 3)
    per_atom: NDArray[np.float64] | None # (F, A)
    per_atom_components: NDArray[np.float64] | None # (F, A, 3)
    per_atom_indices: NDArray[np.int64] | None

    normalization: str
    integrated_weight_before: float
    integrated_weight_after: float
    target_weight: float | None
    source_estimator: str
    weighting: str
    density_units: str
    metadata: Mapping[str, Any]
    signature: DynamicsInputSignature | None
```

7 Input contract

The input must satisfy the `VelocitySpectrumResult` constructor invariants and must record

```
spectral_sidedness == "one_sided"
spectral_scaling == "density"
```

The frequency grid must contain at least two uniformly spaced bins. VS3 accepts both VACF-transform and later Welch source estimators.

7.1 Source weighting

Mass weighting is preferred when the result is interpreted as a vibrational DOS. Uniform or explicit-uniform weighting remains numerically valid, but the metadata records that the result is a velocity-derived VDOS rather than silently declaring it a harmonic phonon DOS.

8 Frequency selection

When `minimum_frequency_thz` is not `None`, VS3 retains existing bins satisfying

$$f_m \geq f_{\min}.$$

It does not interpolate a new boundary bin. The retained grid must contain at least two bins. The threshold and first retained source index are recorded in metadata.

This option can remove a DC or low-frequency diffusive contribution, but the operation changes the normalized object and must remain explicit.

9 Negative-value policy

Diagonal velocity spectra and VDOS projections should be nonnegative in the ideal infinite-sampling limit. A finite transformed VACF may contain negative lobes.

For each real projection, define a scale-aware threshold

$$\epsilon_P = \epsilon_{\text{user}} \max(1, \max |P|).$$

- `negative_policy="error"` rejects any negative sample.
- `negative_policy="clip_roundoff"` clips samples in $[-\epsilon_P, 0)$ and rejects values below $-\epsilon_P$.

Material negative weight is never silently discarded. After permitted clipping, total and per-atom scalar arrays are recomputed from their Cartesian components so exact trace identities are restored.

10 Normalization modes

10.1 none

No scalar area normalization is applied. The retained, validated spectrum is returned with its source density units.

10.2 unit_area

The target is

$$I_{\text{target}} = 1.$$

The result has density units 1/THz.

10.3 degrees_of_freedom

The caller must supply a finite, strictly positive `target_degrees_of_freedom`. Examples include $3N$, $3N - 3$, or a user-defined projected count. VS3 does not guess translational, rotational, rigid-body, or constraint removals.

The density units are `degrees_of_freedom/THz`.

Supplying `target_degrees_of_freedom` for another normalization mode is an error rather than an ignored parameter.

11 Discrete normalization measure

For uniform one-sided FFT bins,

$$I = \Delta f \sum_m g_m.$$

VS3 calls `spectral_bin_integral`. It does not call `numpy.trapezoid`, SciPy quadrature, or an interpolating integrator. The one-sided endpoint weights were already handled by the spectral estimator.

Let I_0 be the retained pre-normalization weight and I_* the target. For normalized modes,

$$a = \frac{I_*}{I_0}, \quad g'_m = ag_m.$$

Exactly the same scalar a is applied to total, Cartesian, and per-atom projections.

12 Algorithm

1. Validate the result type and one-sided density metadata.
2. Validate the requested normalization and tolerance.
3. Resolve the existing-bin frequency slice.
4. Copy total, component, and per-atom projections on that slice.
5. Apply the explicit negative policy to component projections.
6. Recompute scalar projections from Cartesian traces.
7. Compute the retained total weight with `spectral_bin_integral`.
8. Reject nonpositive or nonfinite retained weight.
9. Resolve the target and one scalar normalization factor.
10. Scale every retained projection by the same factor.
11. Recompute the post-normalization bin measure.
12. Construct `VDOSResult` with source and operation provenance.

13 Result identities

The constructor validates

$$g(f) = g_x(f) + g_y(f) + g_z(f),$$

and, when per-atom components exist,

$$g_i(f) = g_{ix}(f) + g_{iy}(f) + g_{iz}(f).$$

For `unit_area`,

$$\Delta f \sum_m g_m = 1.$$

For degrees-of-freedom normalization,

$$\Delta f \sum_m g_m = N_{\text{dof,target}}.$$

14 Edge cases and failure policy

VS3 rejects:

- non-`VelocitySpectrumResult` input;
- unsupported spectral sidedness or scaling;
- fewer than two retained frequency bins;
- invalid normalization or negative policy;
- a missing, nonfinite, or nonpositive degrees-of-freedom target;
- a target supplied for another normalization mode;
- a nonfinite or negative frequency threshold;
- nonfinite or negative tolerance;
- material negative total, Cartesian, or per-atom spectral values;
- zero or negative retained total weight.

No warning substitutes for an invalid normalization.

15 Complexity and memory

For F retained frequency bins and A stored per-atom projections, time and additional memory are

$$O(FA)$$

when per-atom data exist, and $O(F)$ otherwise. The source result is not modified.

16 Required tests

- unit-area normalization through the discrete bin measure;
- explicit $3N$, $3N - 3$, and arbitrary target normalization;
- exact preservation for `normalization="none"` without cropping or clipping;
- total/component and per-atom/component trace identities;
- zero-padding invariance of normalized integrated weight;
- explicit low-frequency existing-bin cropping;
- rejection of a threshold that leaves fewer than two bins;
- material-negative rejection;
- roundoff-only clipping;
- no mutation of the source result;
- deliberate inequality between the normative bin sum and trapezoidal endpoint weighting;
- result-constructor shape, unit, and metadata validation;
- top-level public import smoke testing.

17 Deferred work

VS3 does not implement:

- direct Welch estimation;
- spectrum or VDOS plotting;
- peak finding or mode assignment;
- quantum correction factors;
- harmonic dynamical-matrix phonon DOS;
- 2PT entropy or free energy;
- uncertainty estimation.

18 H0 signature propagation and deep immutability

A `VDOSResult` preserves the input `VelocitySpectrumResult.signature`. VDOS normalization changes the spectral measure but not the trajectory, atom selection, drift, velocity source, or analysis subspace identity.

Every frequency/projection array is owned and read-only, and metadata is recursively immutable. A supplied signature must retain the full Cartesian source subspace. The common contract is `docs/specs/analysis/_dynamics_common_spec`

19 References

- [1] A. Rahman, “Correlations in the Motion of Atoms in Liquid Argon,” *Physical Review* **136**, A405-A411 (1964). DOI: 10.1103/PhysRev.136.A405.
- [2] S.-T. Lin, M. Blanco, and W. A. Goddard III, “The Two-Phase Model for Calculating Thermodynamic Properties of Liquids from Molecular Dynamics: Validation for the Phase Diagram of Lennard-Jones Fluids,” *Journal of Chemical Physics* **119**, 11792-11805 (2003). DOI: 10.1063/1.1624057.